

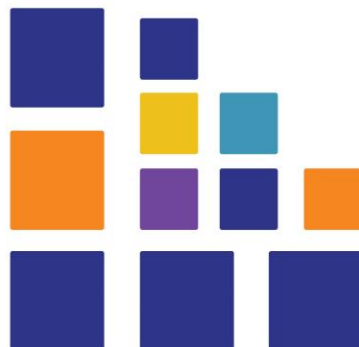
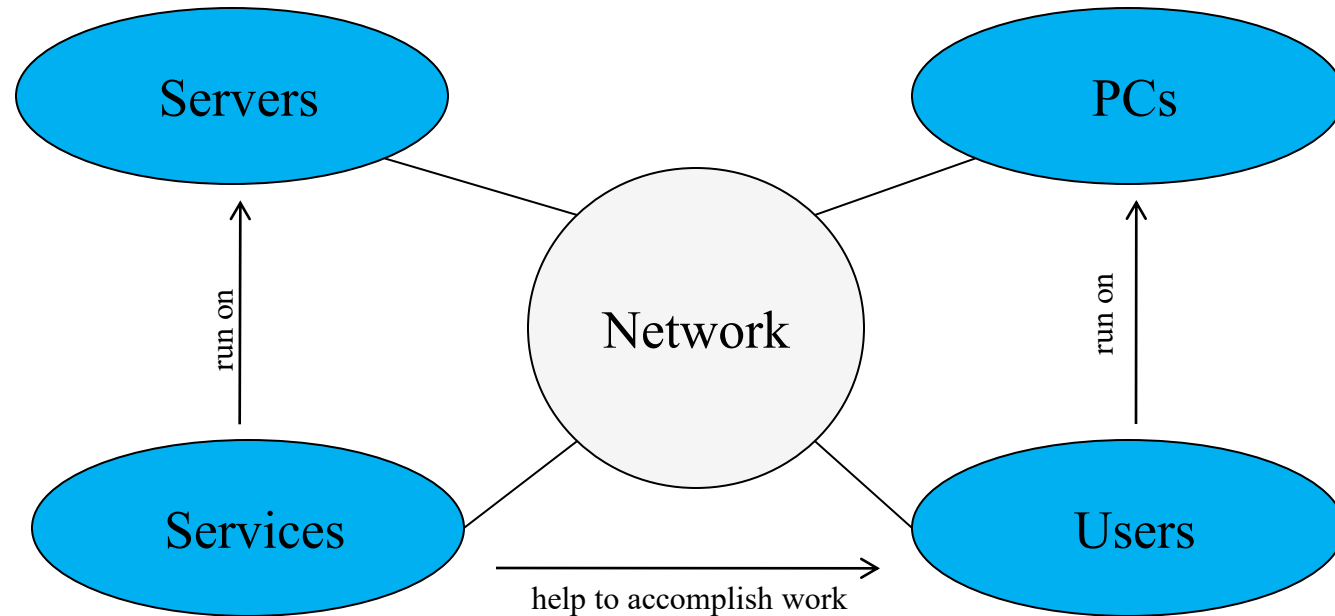


Chapter 1:

Introduction to System Administration and OS Installation

What is a system?

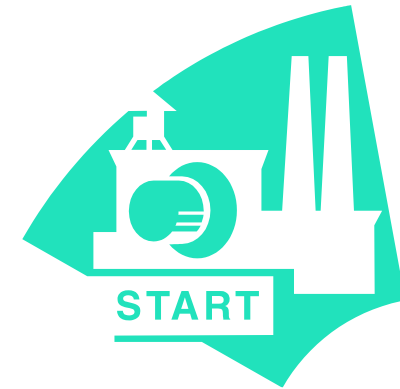
System: An organized collection of computers interacting with a group of users.



What is a system administrator? / Why we need System Administrator?

System Admin Tasks and Duties:

- install systems (including clients/servers): hardware, software and os.
- upgrade systems: hardware, software and os.
- backups
- start/stop system (reboot)



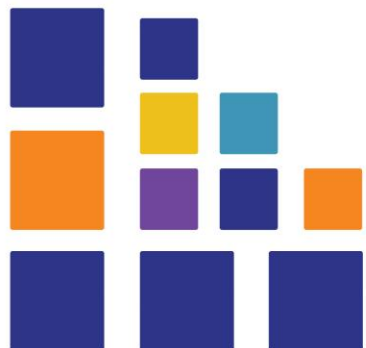
Other Duties

- create accounts (add/delete), passwords, account management
- job scheduling
- security
- performance monitoring and tuning

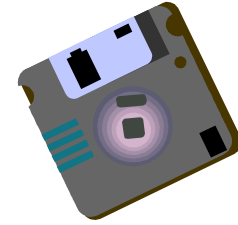


Some More Duties

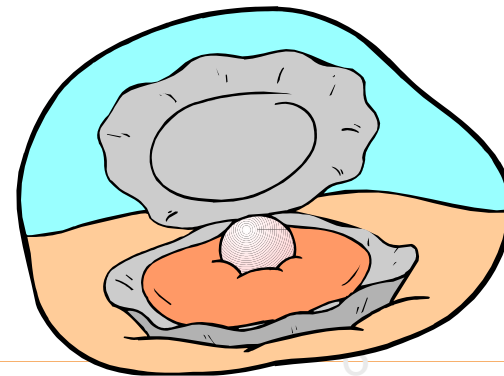
- documentation and testing
- communication and help (with user community and internal)
- networking (client/server)
- handling printers, tape drives, modems, disks, UPS, and other peripherals
- Memory Management



Almost The Last Duties



- disk space management (formatting, partitions, quotas)
- writing/modifying scripts (perl, shell, C, etc..)
- running specialized services/servers (email server, web server, DNS etc..)
- problem resolution

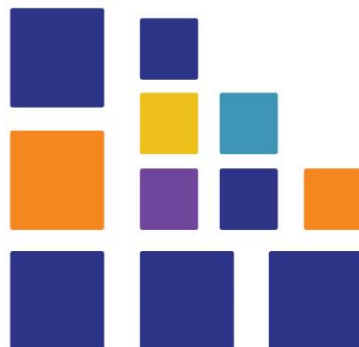


Last Duties

- Training (others and oneself)
- Fixing Bugs
- Automate Tasks
- Maintain system files
- Analysis of logs/systems, collect stats, reports
- Planning and Recommendations
- Work with vendors, customers
- Research New Technologies

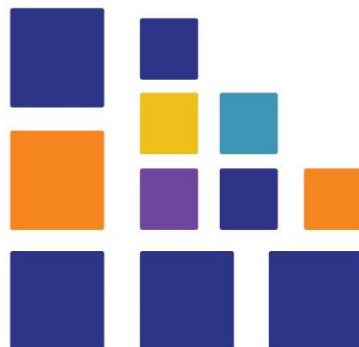


Linux Features



Free

- Freedom of Software
- Is Linux free of cost?



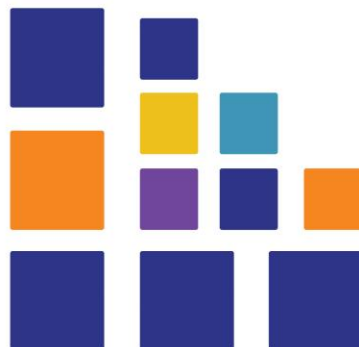
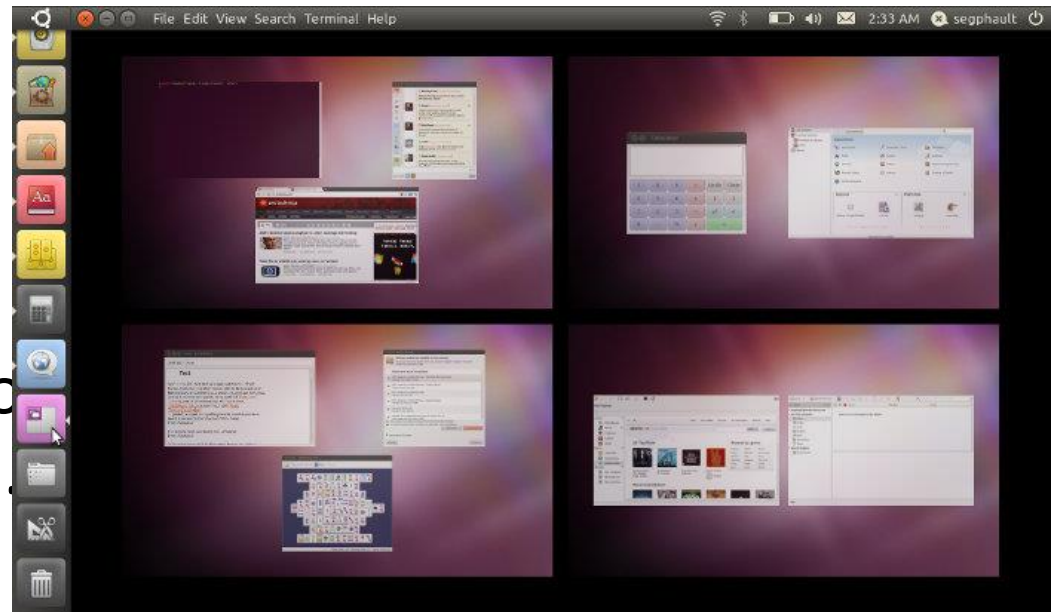
Live CD

- To be able to load a complete desktop on any computer by booting it from a CD or USB.



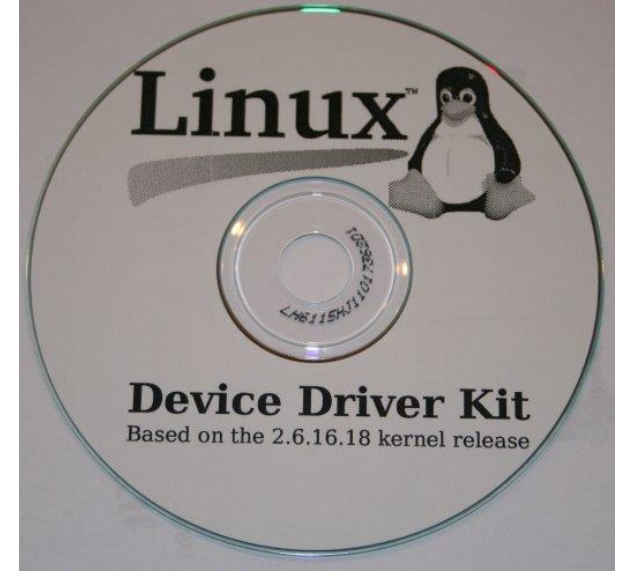
Multiple Desktops

- you could have your email and internet on one desktop, a game
- A flick of the mouse is all that's required to switch between them.



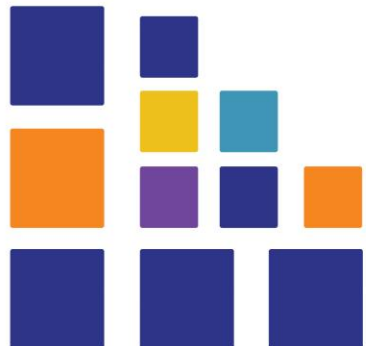
Drivers

- most hardware requires you to install drivers from the manufacturer.
- the majority of hardware is supported directly by the Linux kernel
- just plug and play – this is one of the reasons live CDs work so well.



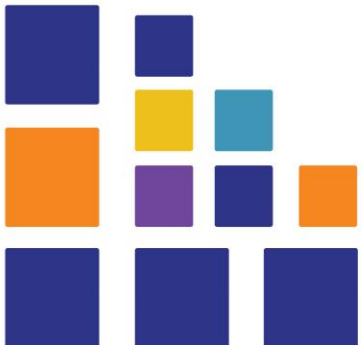
Platform Independence

- Linux runs on just about any hardware you can name, from mobile phones to supercomputers.
- The open source nature of the kernel and software means it can be ported to architecture by a third party
- The end user doesn't need to care about the underlying hardware



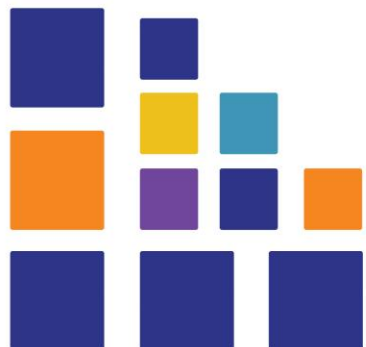
Interoperability

- Linux plays well with other systems.
- It recognizes that there's a place for Windows and Mac OS X and will install alongside them, share files with them, and generally be nice to them.



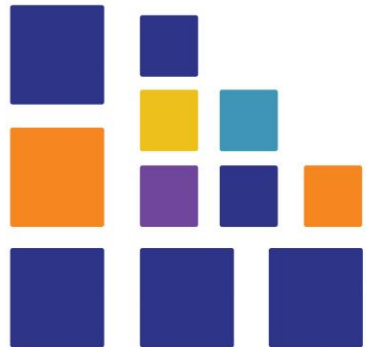
Security

- Linux is inherently more secure than Windows, because security is a feature of the core system,
- With a firewall at the heart of the kernel and the virtual impossibility of slipping malware into the software repositories,
- Linux don't use much of security programs from third party



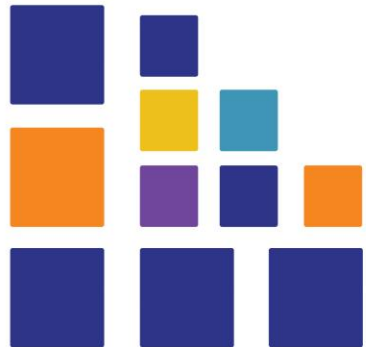
Reinstallations

- When problems occur in Linux, they can be fixed directly. You don't even need to reinstall when you want to update to the latest version of your distribution.



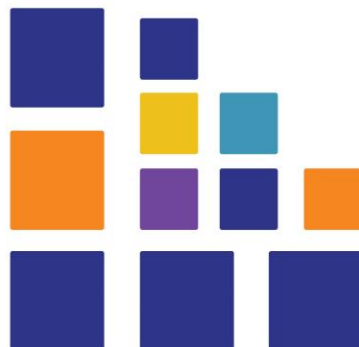
Reinstallations

- When problems occur in Linux, they can be fixed directly. You don't even need to reinstall when you want to update to the latest version of your distribution.



Linux Installation

Download CentOS 8 ISO installation image from the official website of CentOS. Visit the official website of CentOS and click on CentOS Linux DVD ISO button as marked in the screenshot below.



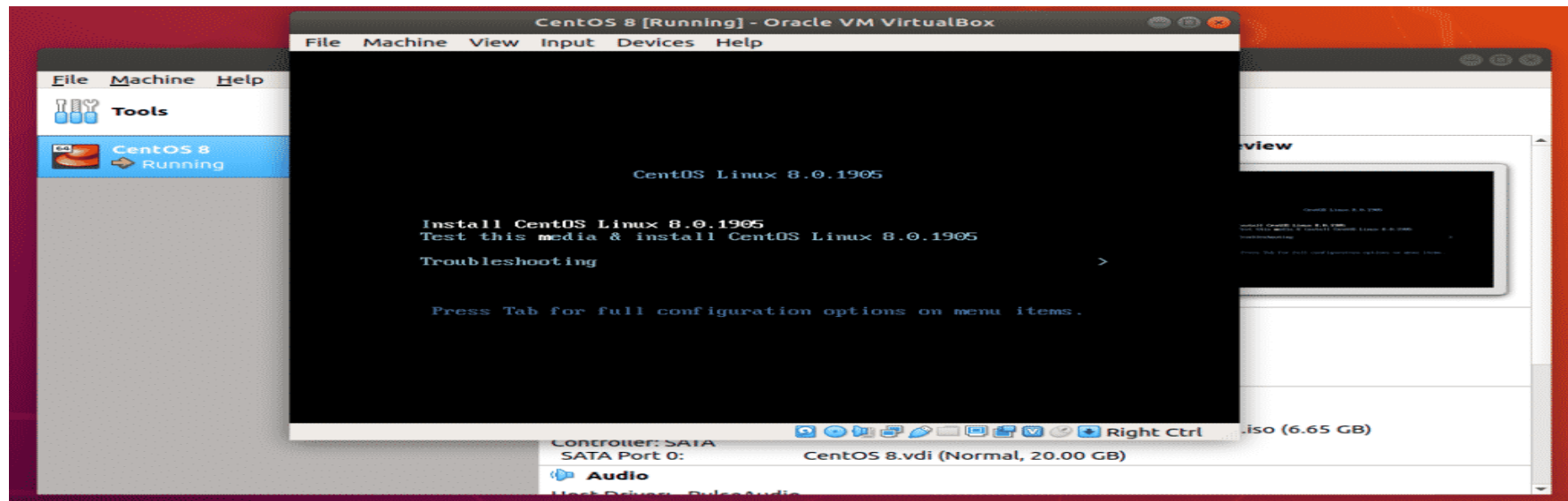
Step 1:

Select the VM and click on Start.



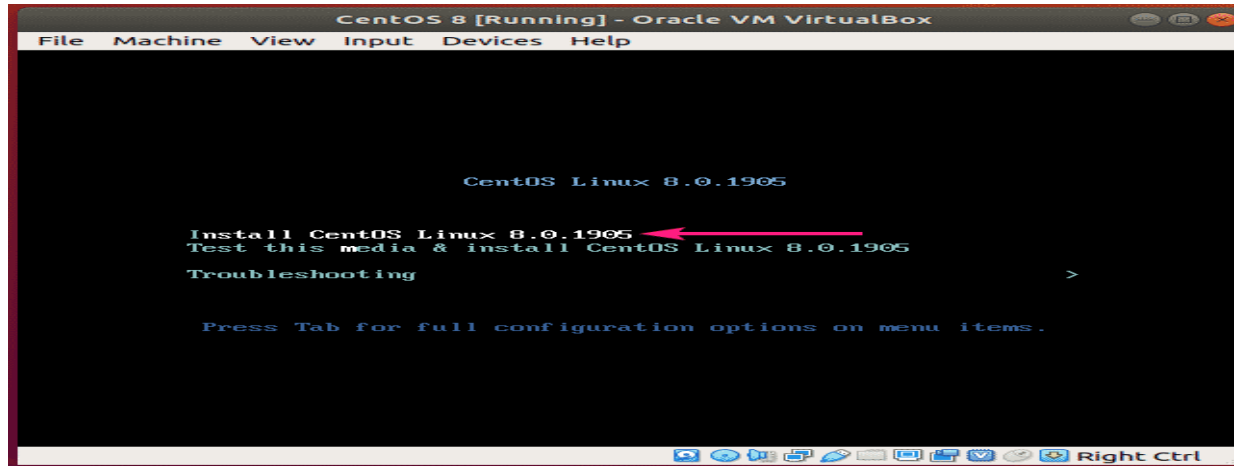
Step 2:

The VM should start and you should see the GRUB boot menu as you can see in the screenshot below.



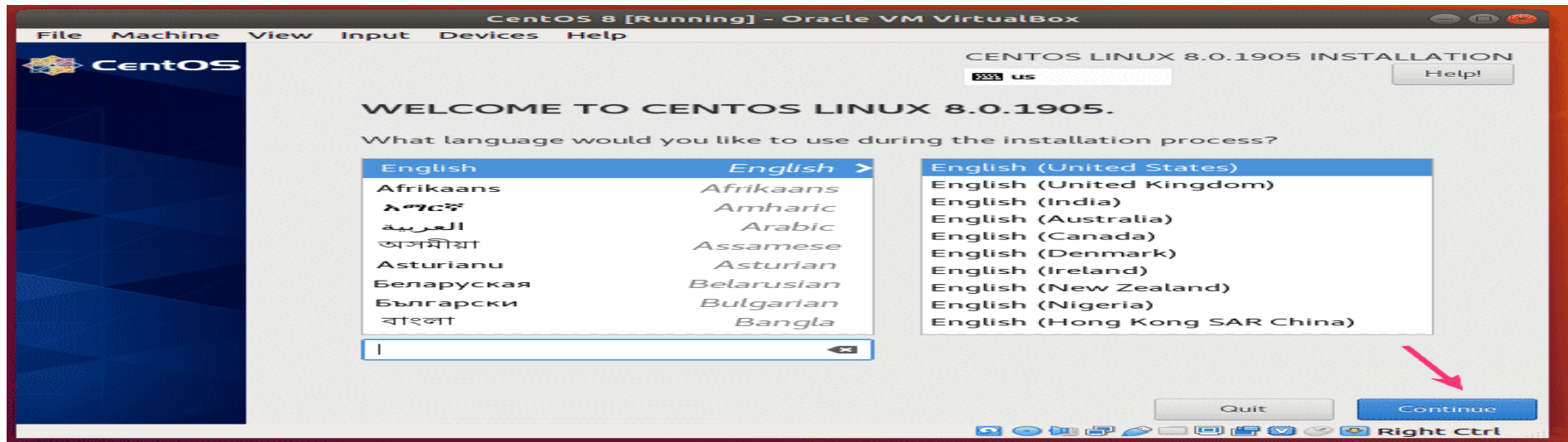
Step 3:

Now, select Install CentOS Linux 8.0.1905 from the GRUB menu and press <Enter>.

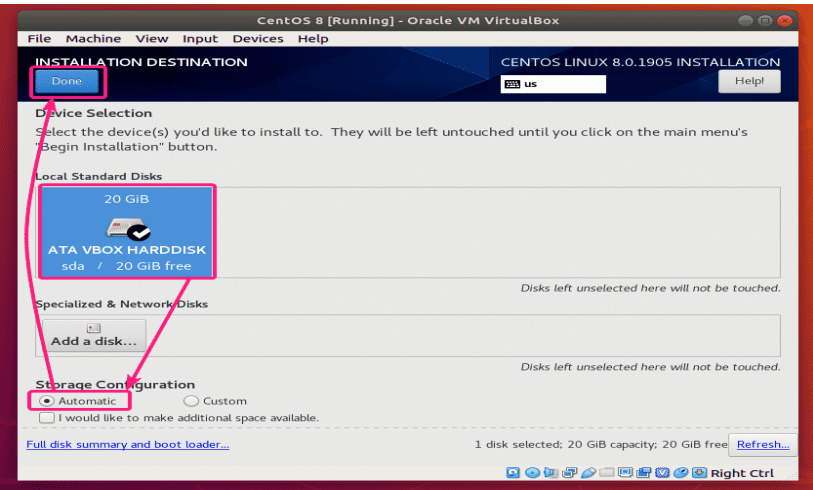
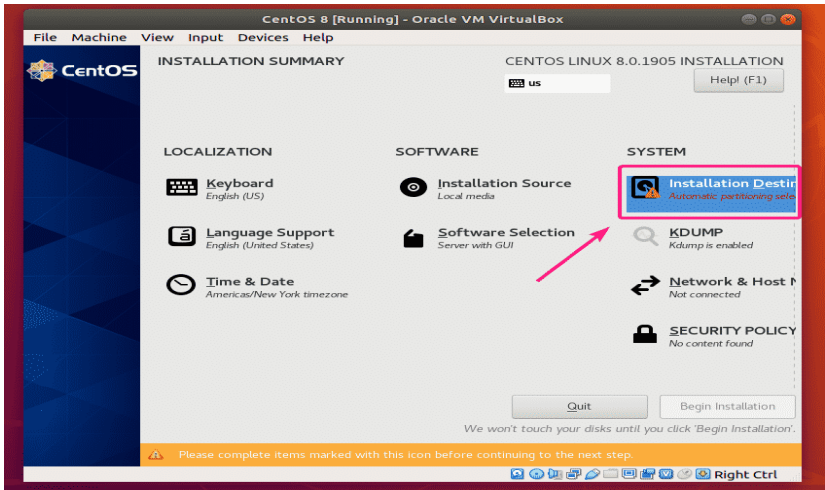


Step 4:

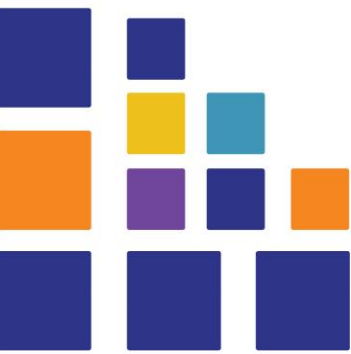
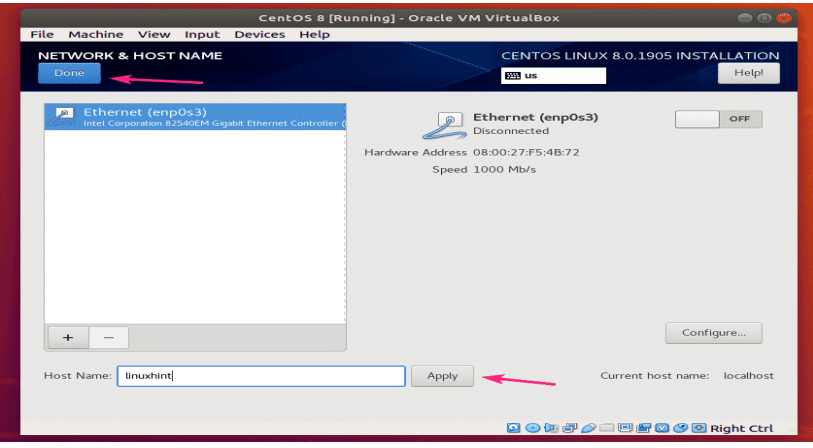
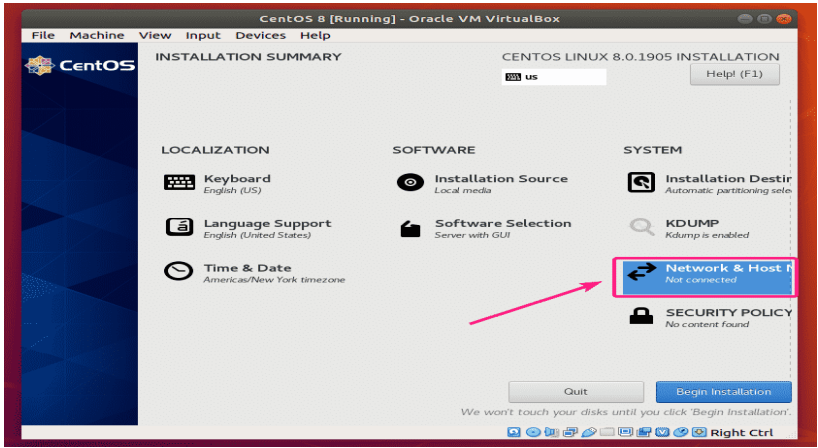
The CentOS 8 installer should work. Now, you can install CentOS 8 as usual on the virtual machine. Select your language and click on Continue.



Step 5:
Now, click on Installation Destination. select the virtual hard drive, select Automatic from Storage Configuration section and click on Done.



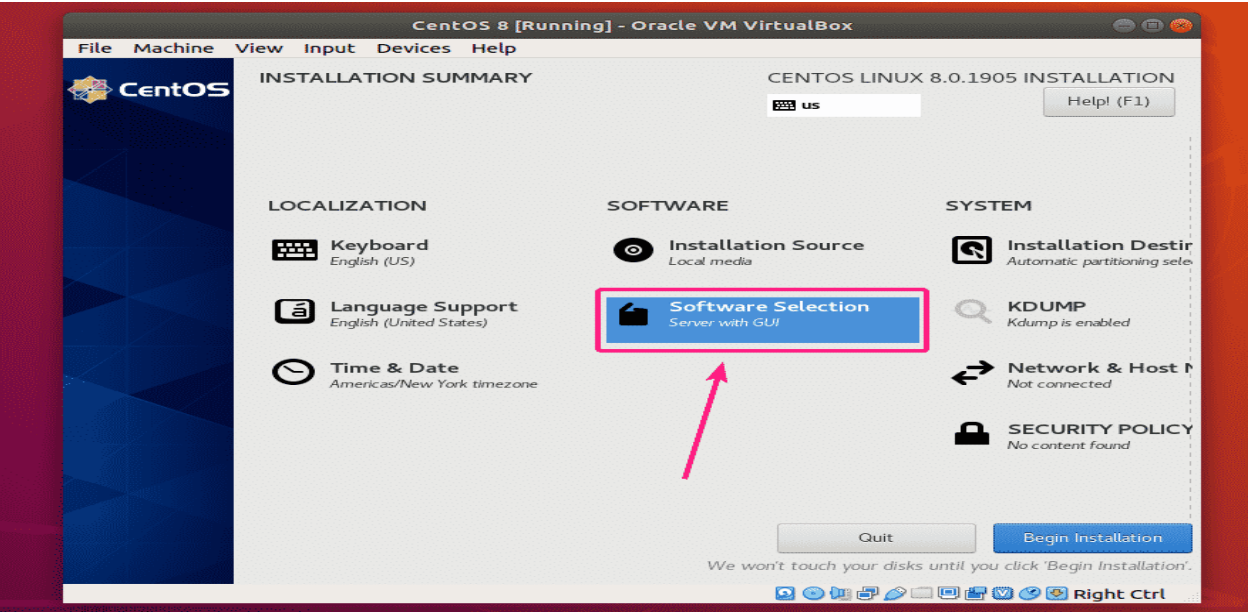
Step 6:
Now, click on Network & Host Name. Type in a host name and click on Apply. Then, click on Done.



Step 7:

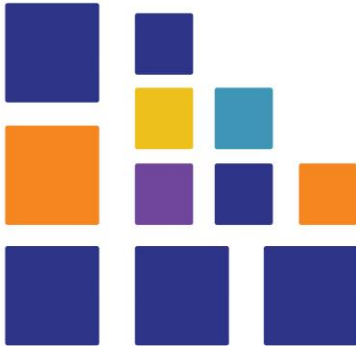
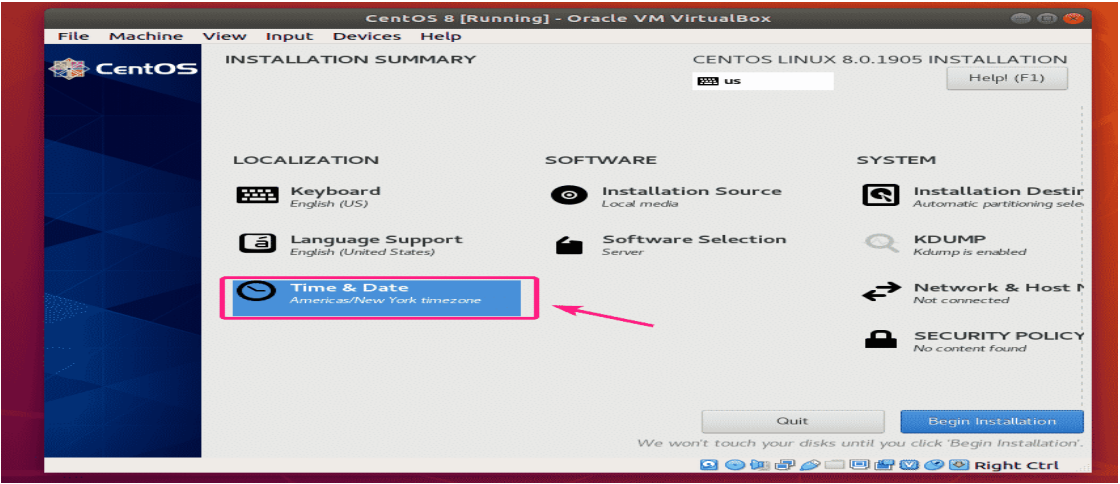
If you want to install CentOS 8 server with graphical user interface, then you don't have to do anything else.

[You can select here server/client]



Step 8:

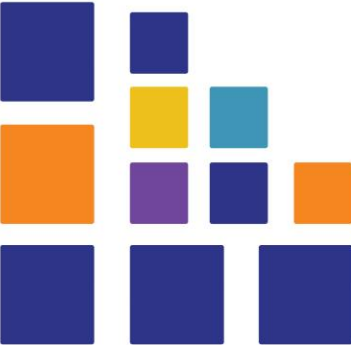
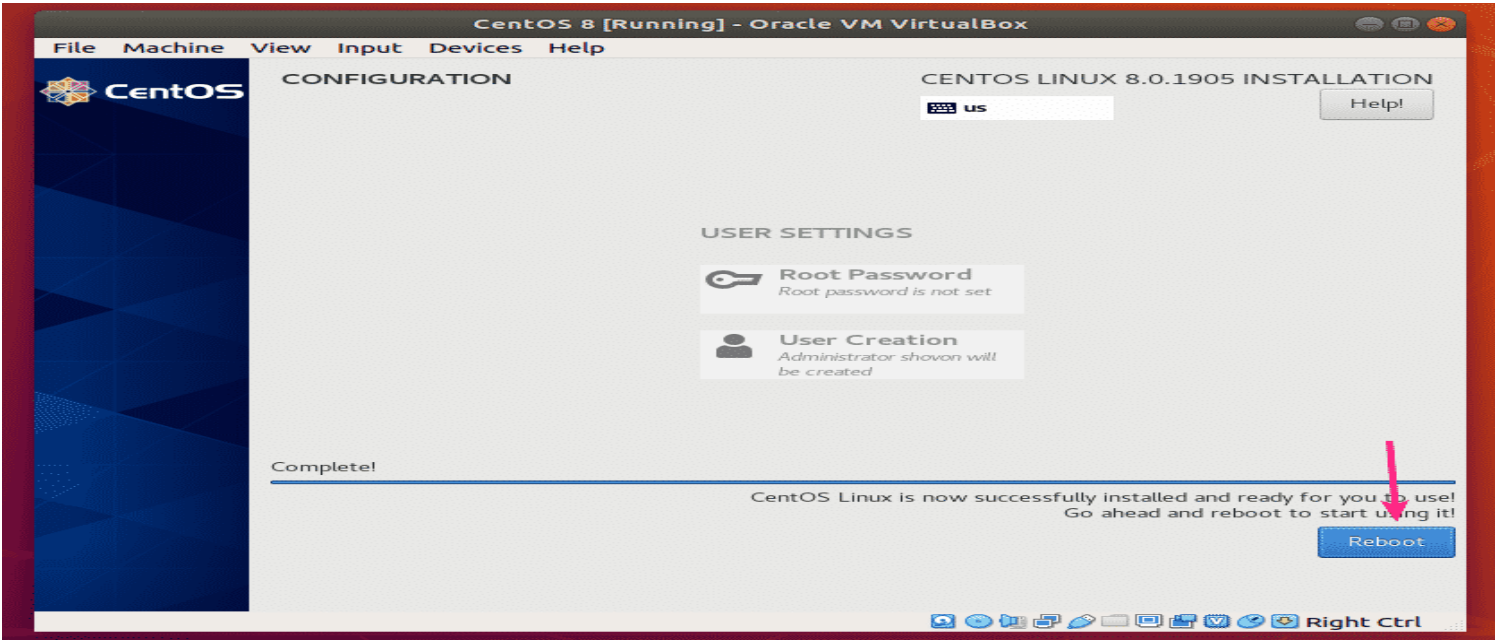
If you want to set up your time zone, click on Time & Date.



Step 9:
click on Begin Installation.



Step 10:
you have to create a root password and new login user, then reboot the system.



Now you are getting the CentOS login Screen



End of Chapter 1

